

The local limit of directed animals on the square lattice

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joint work with **E. Maurel-Ségala** and **A. Singh**



Discrete Random Structures II

Bedlewo, May 2026

Directed animals on the square lattice

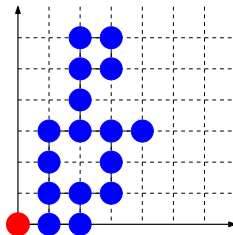
View $\mathbb{Z}^2$ as a directed graph with North/East edges.

Directed animal on $\mathbb{Z}^2$

A directed animal (D.A.) with source set $S \subset \mathbb{Z}^2$ is a finite set $\mathbf{A} \subset \mathbb{Z}^2$ such that:

- 1 $S \subset \mathbf{A}$.
- 2 For each vertex $v \in \mathbf{A}$, there is a vertex $u \in S$ and a North/East path from u to v entirely contained in $\mathbf{A}$.

An example with **source set** the origin, also called a *pyramid*.



This is a directed animal

Directed animals on the square lattice

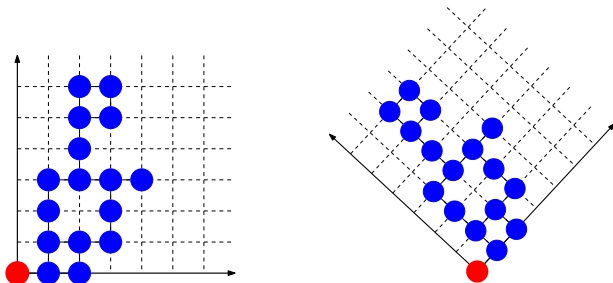
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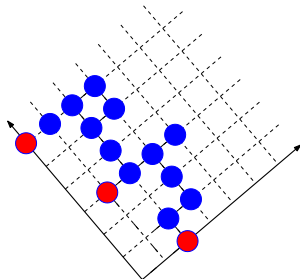
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Another example with a source set of size 3.



Why directed animals ?

- Non-directed animals too hard.
- Directed animals form a *partially* solvable model (partially = enumeration by volume only, essentially)
- More complicated than trees, yet *do* hide trees.

(Very incomplete) bibliography

1 Study of D.A. via hard sphere gas model

► Dhar (1982); Dhar Phani Barna (1982); Nadal (1982); Derrida Nadal Vannimenus (1982), Hakim Nadal (1983), Dhar (1983); Bousquet-Mélou, Conway (1996); Bousquet-Melou (1998); Le Borgne, Marckert (2007); Albenque (2009)

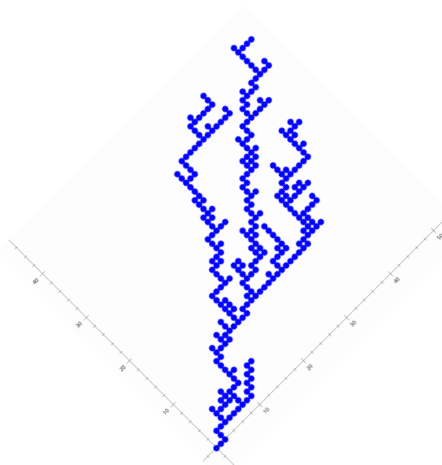
2 Study of D.A. via heap of pieces → bijections with trees

► Viennot (1984, 1986); Betrema, Penaud (1993); Corteel, Denise and Gouyou-Beauchamps (2000); Bacher (2009)

Goal of the talk

What does a large uniformly sampled random D.A. look like around the origin ?

► **We study the local limit of D.A. rooted at the origin.**
(probabilistic vs combinatorial approach)



1. Enumeration by volume

- ▶ Dhar method: a *dual* solvable gas model
- ▶ Viennot *heaps of pieces* decomposition

2. Local limits of large random directed animals

- ▶ Heaps of pieces describing a *random walk*
- ▶ Markov structure of the *layers* and sausaging

Dhar's key observation (82) :

On a general oriented graph, there is a duality between D.A. and a certain hard-particle gas model on the same graph.

- Let $G = (V, \vec{E})$ be an *oriented* graph.
- Define directed animals with source set $S \subset V$ on G as before, by replacing $\mathbb{Z}^2$ by G and North/East paths by directed paths.
- The (*volume*) *generating function* of D.A. with source set S is

$$G_S(t) = \sum_{A \text{ D.A. with source } S} t^{|A|}.$$

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A gas is a random subset of the vertices of the graph, with the constraint that adjacent vertices cannot both be occupied.

- On a *finite* graph, the gas model is defined recursively *from the leaves downwards* by

$$X_v = \begin{cases} B_v, & \text{if } v \text{ is a leaf,} \\ B_v \prod_{w: vw \in \vec{E}} (1 - X_w), & \text{otherwise,} \end{cases}$$

where $(B_v)_{v \in V}$ are independent Bernoulli variables with parameter p .

- The definition also extends to an *infinite* graph when p is small enough (below the radius of convergence of G_u for each u), see [\[Le Borgne, Marckert, 2007\]](#).

Dhar duality

Dhar's key observation (82) :

On a general oriented graph, there is a duality between D.A. and a certain hard-particle gas model on the same graph.

Assume $G = (V, \vec{E})$ is agreeable (finite degrees, no directed loop, no multiple edges).

Theorem Dhar (1982); Bousquet-Melou (1998); Le Borgne–Marckert (2008)

For every finite free^a subset $S \subset V$, and for every p less than the radius of convergence of G_S ,

$$\mathbb{E}_p \left[\prod_{s \in S} X_s \right] = (-1)^{|S|} G_S(-p).$$

In particular, for $S = \{u\}$

$$\mathbb{E}_p[X_u] = -G_{\{u\}}(-p).$$

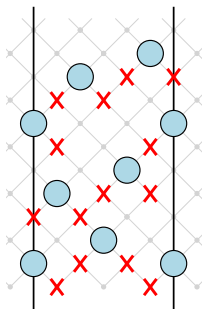
^acontains no two vertices that are ancestors of each other

Computing the mean density for the square lattice

- It remains to compute $\mathbb{E}_p[X_u]$ on the square lattice $\rightarrow$ approximation by **cylinders** of finite width n .
- Layers of the gas model form a **finite state space irreducible** Markov chain; if C stands for the set of **occupied** vertices in a layer, the invariant measure is:

$$\pi_n(C) \propto \left(\frac{p}{1-p} \right)^{|C|} (1-p)^{|[C]|}.$$

with $[C] = (C-1) \cup (C+1)$.



- Allows the computation of the limit of $\mathbb{E}_p[X_u^{(n)}]$, hence of the g.f. of pyramids:

$$\mathbb{E}_p[X_u] = \lim_{n \rightarrow \infty} \mathbb{E}_p[X_u^{(n)}] = \frac{1}{2} \left(1 - \sqrt{\frac{1-p}{1+3p}} \right) = -G(-p)$$

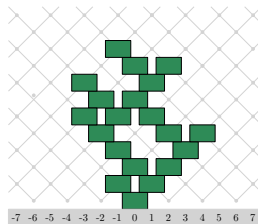
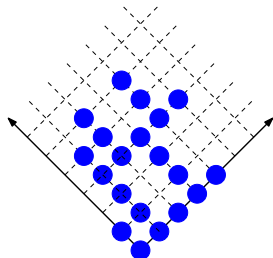
Viennot's heap of pieces

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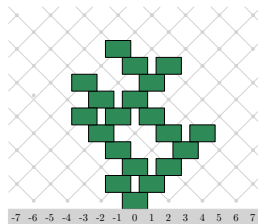
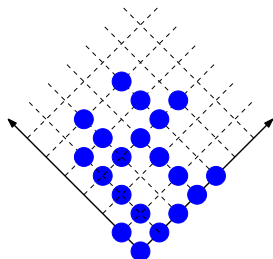
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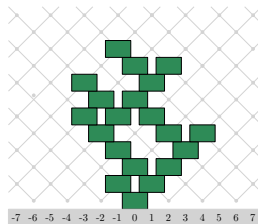
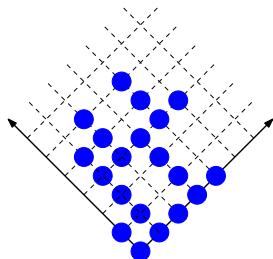
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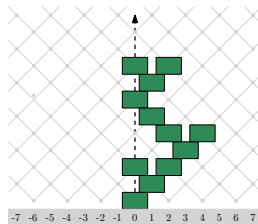
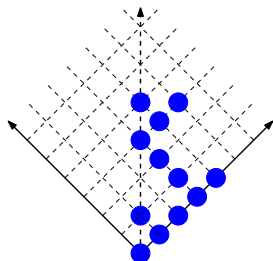


Set of dominoes where one domino is on the floor and every other domino is supported by a domino under it

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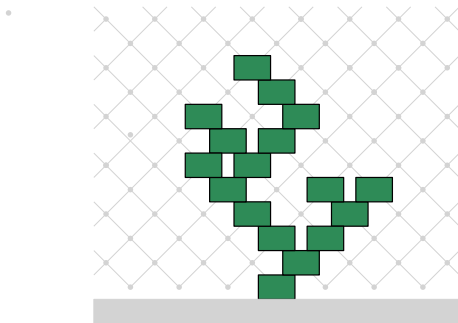


half-pyramid



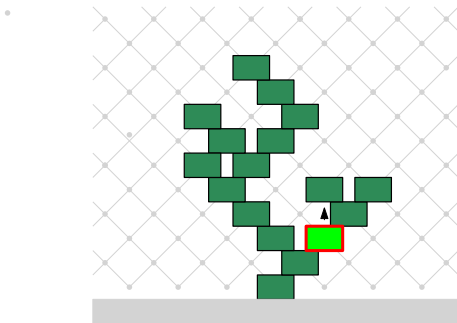
pyramid where all dominoes
have non-negative x-coordinates

Viennot's heap of pieces : the “push-up” operation



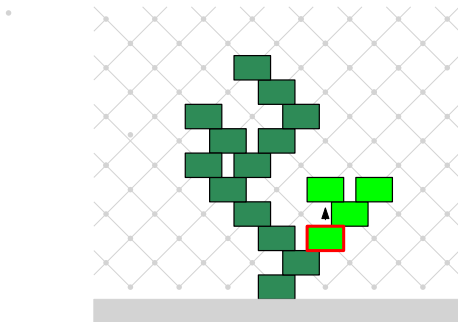
Viennot's heap of pieces : the “push-up” operation

- Lifting up a domino: bring along the pyramid sitting over it.



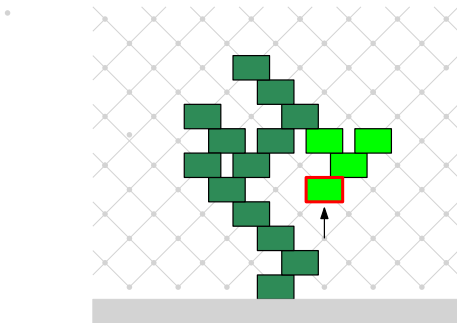
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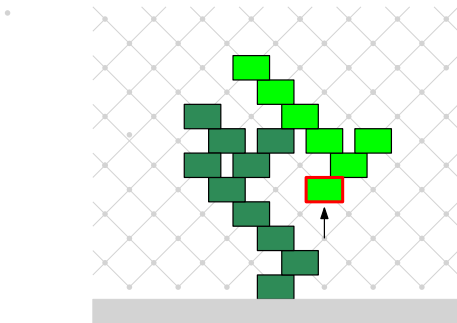
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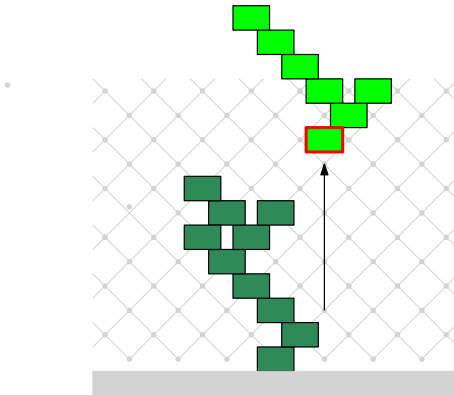
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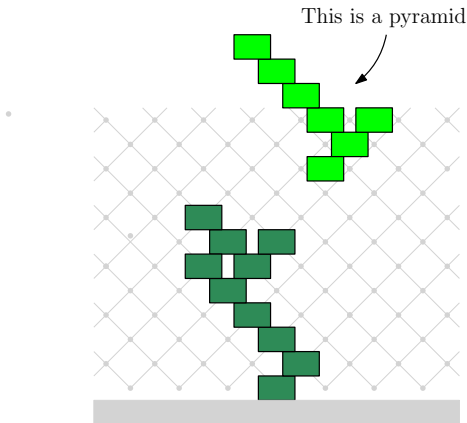
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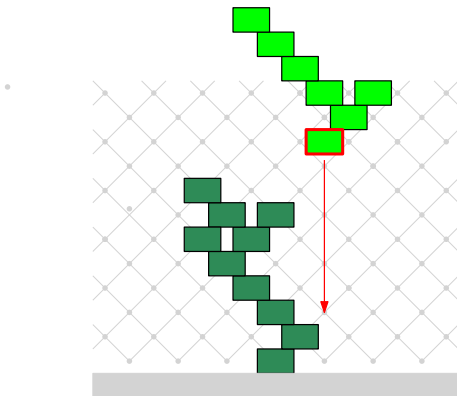
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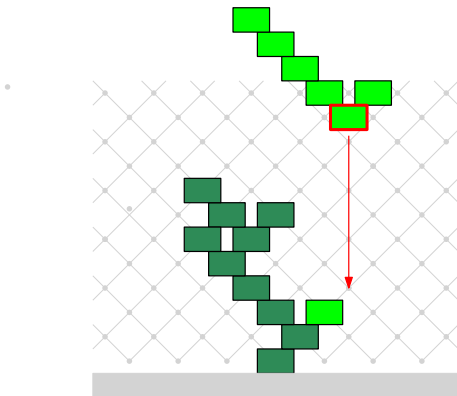
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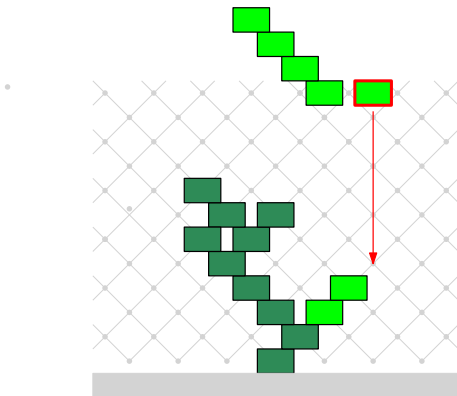
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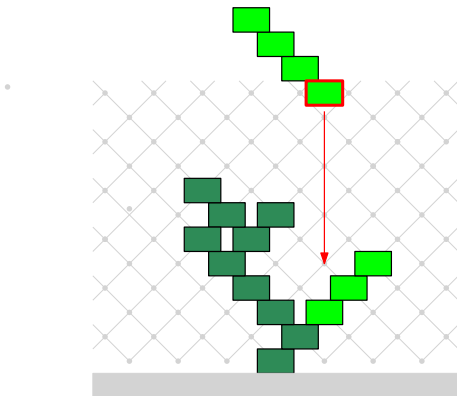
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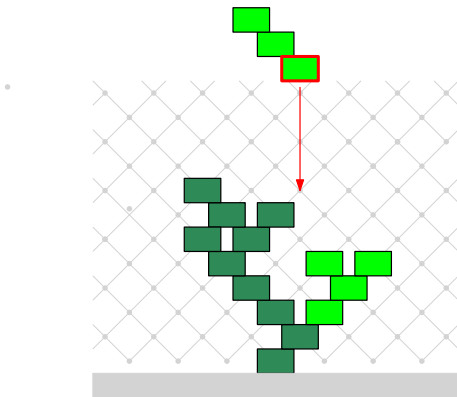
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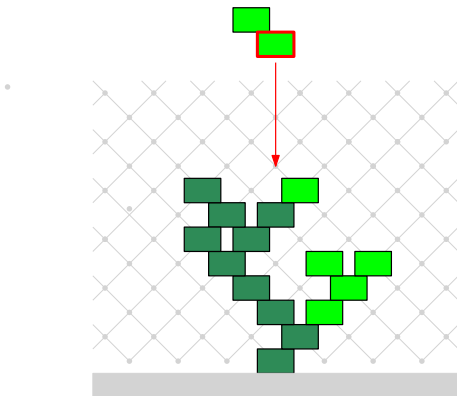
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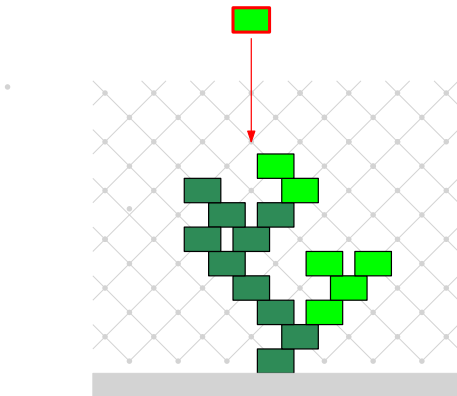
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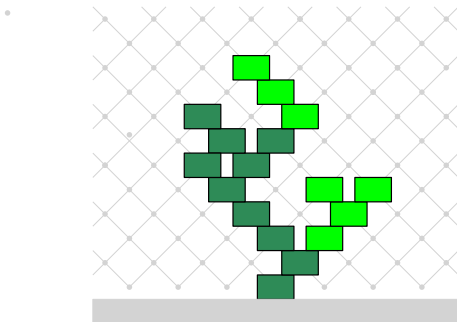
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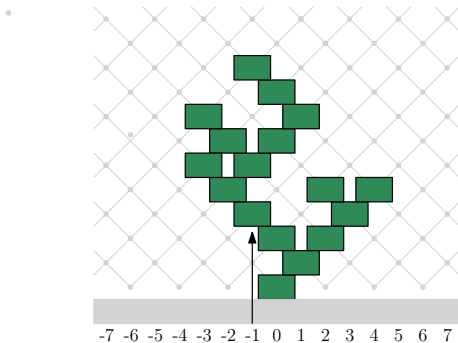
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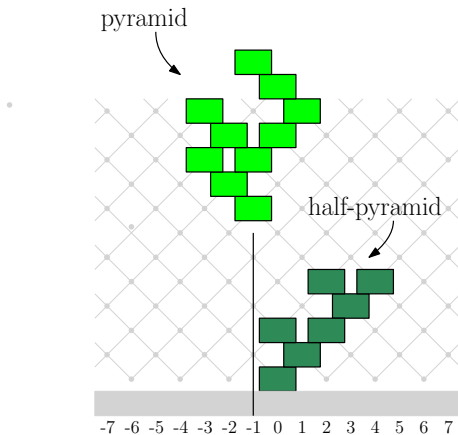
Pyramid / half-pyramid decomposition



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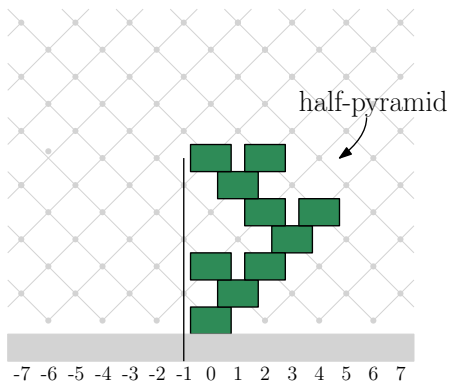
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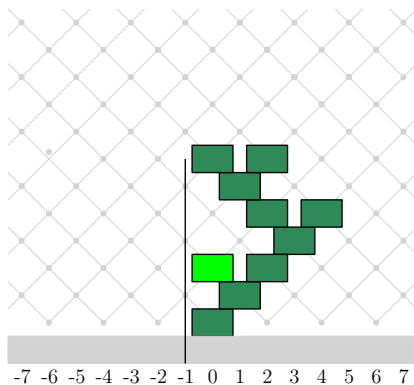
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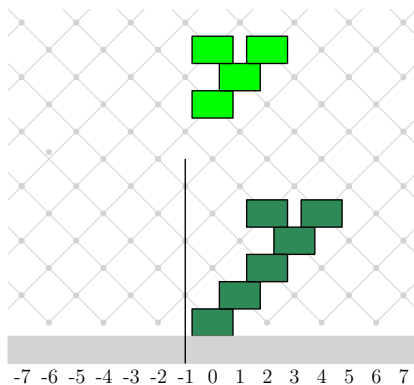
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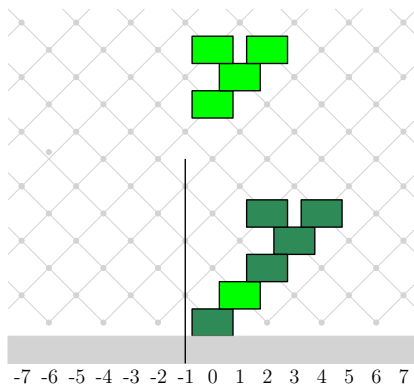
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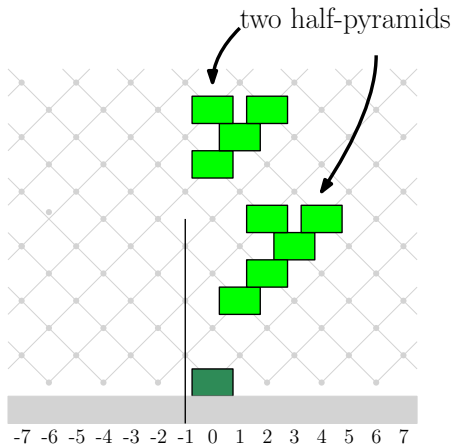
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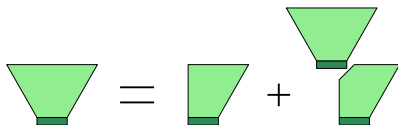


Pyramid decomposition for the g.f.

$$\mathcal{P}(s) = \sum_{\text{pyramids } P} s^{|P|} \quad \text{and} \quad \mathcal{H}(s) = \sum_{\text{half-pyramids } H} s^{|H|}$$

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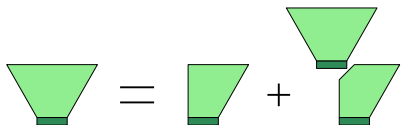
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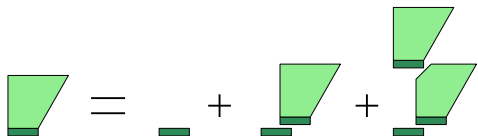
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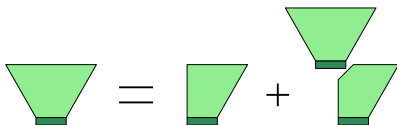
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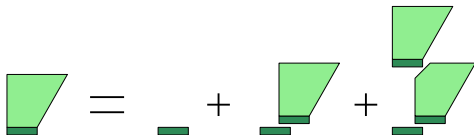
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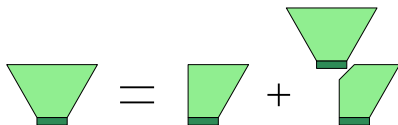
$$\mathcal{P}(s) = \frac{1}{2} \left(\frac{1+s}{1-3s} - 1 \right)$$



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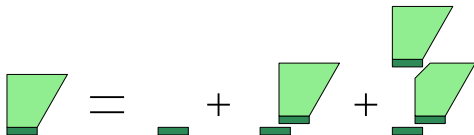
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OEIS A005773

$$\mathcal{P}(s) = \frac{1}{2} \left(\frac{1+s}{1-3s} - 1 \right)$$

$$[\mathcal{P}]_n \sim C \frac{3^n}{\sqrt{n}}$$



OEIS A001006 (Motzkin)

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► We now turn to the local limits of large random directed animals.

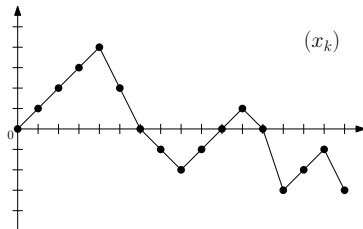
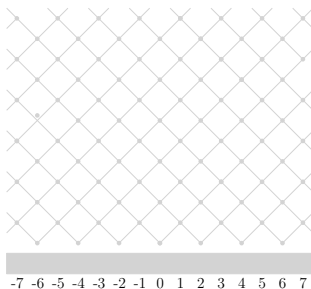
Path encoding of a directed animal

Proposition

Pyramids are in **bijection** with paths (x_k) such that

- 1 $x_0 = 0$ and $x_{k+1} - x_k \in \mathbb{Z}_-^* \cup \{1\}$.
- 2 $x_k \geq \min_{i < k} x_i - 1$ i.e. x never falls below its current infimum by more than 1.

Example.



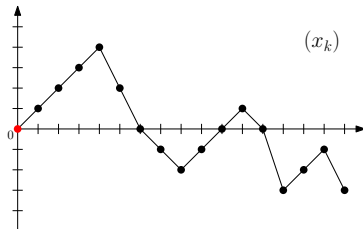
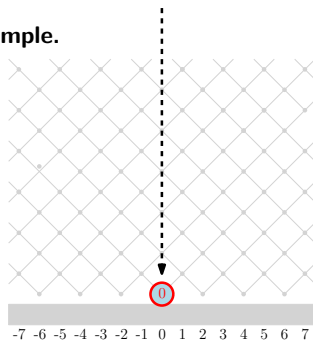
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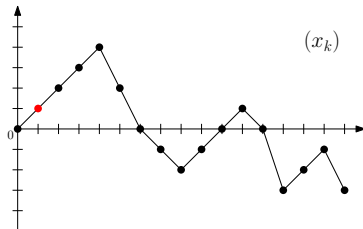
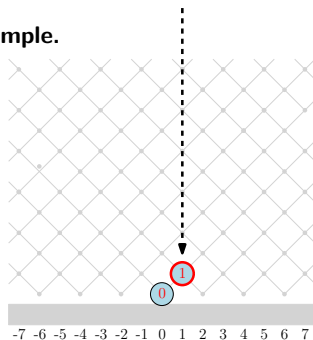
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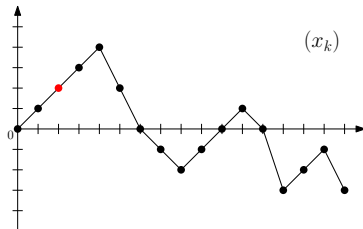
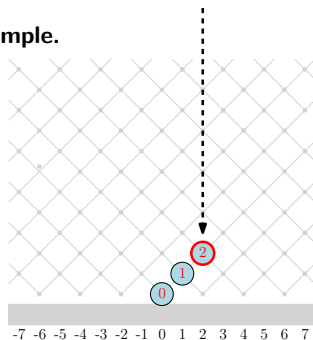
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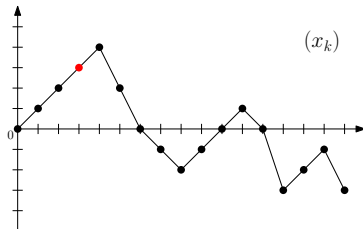
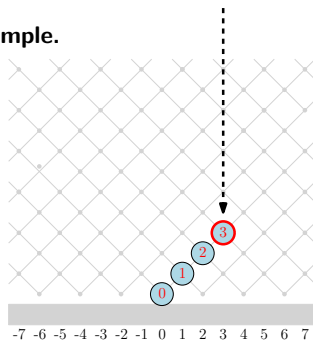
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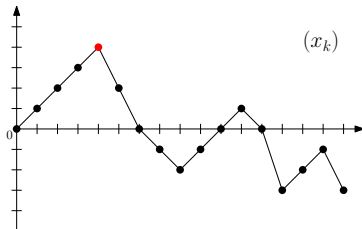
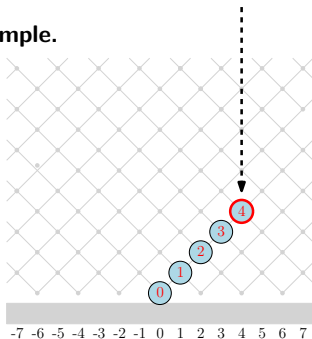
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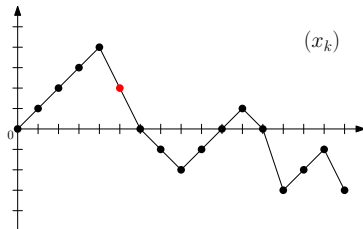
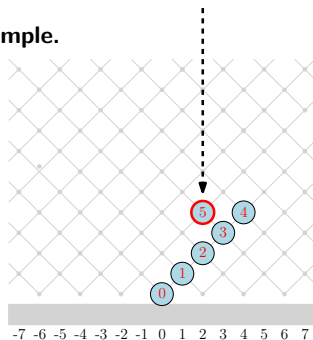
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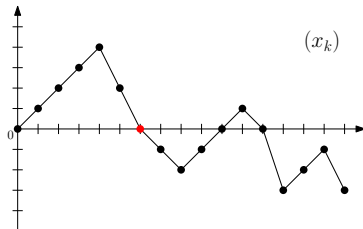
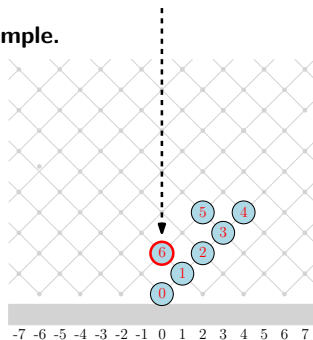
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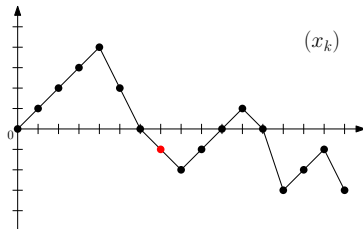
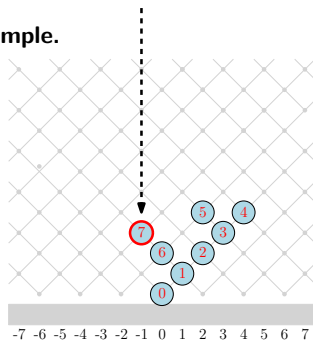
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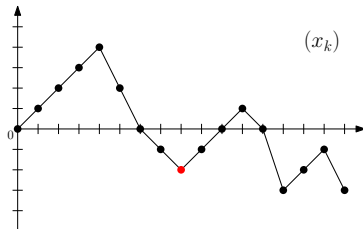
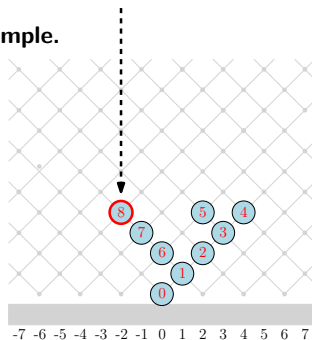
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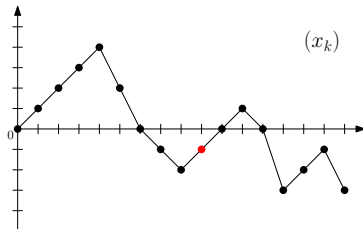
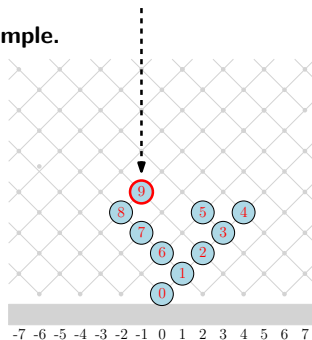
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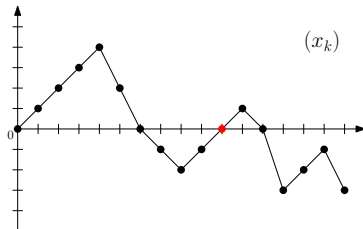
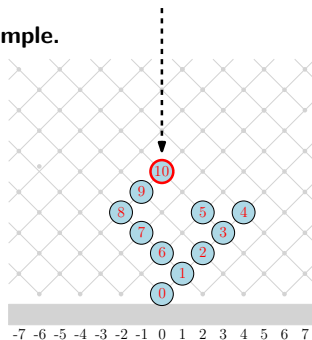
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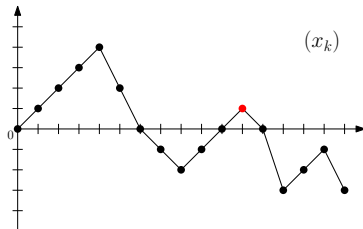
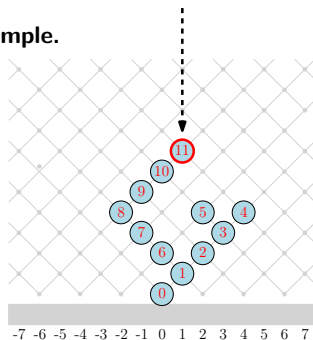
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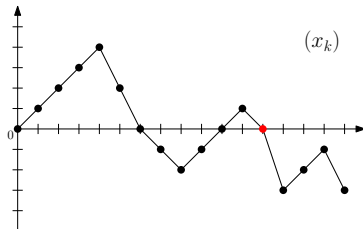
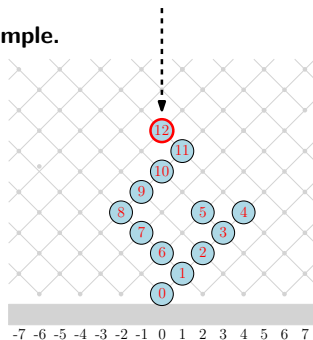
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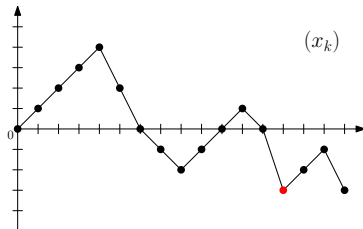
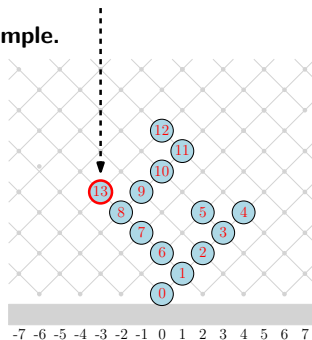
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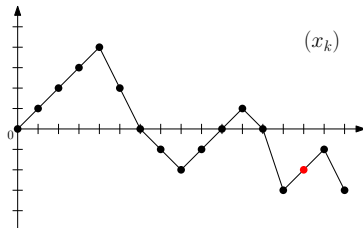
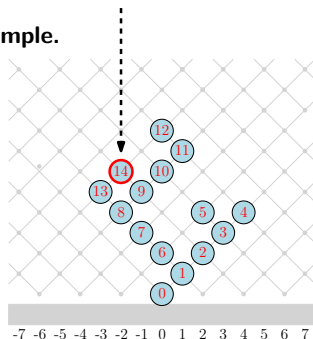
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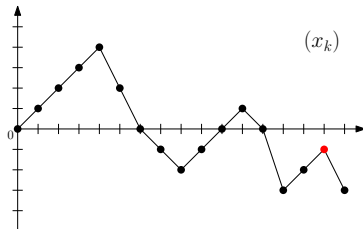
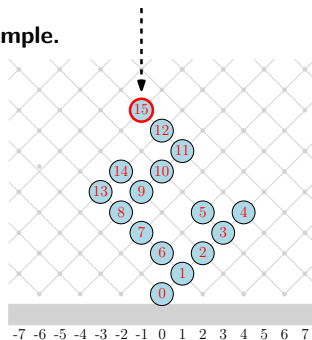
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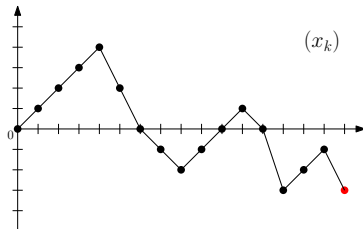
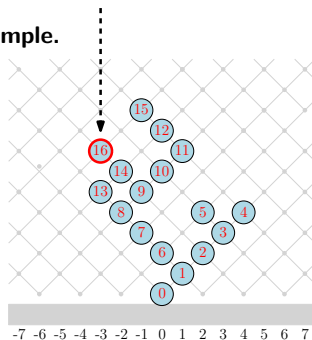
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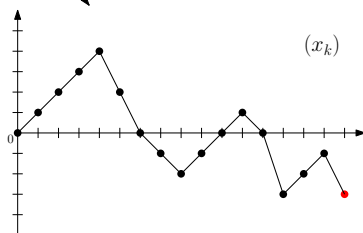
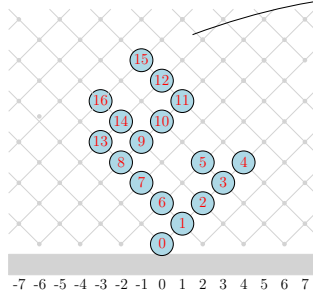
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Example.

Reconstruction of the path from the ordering of the vertices



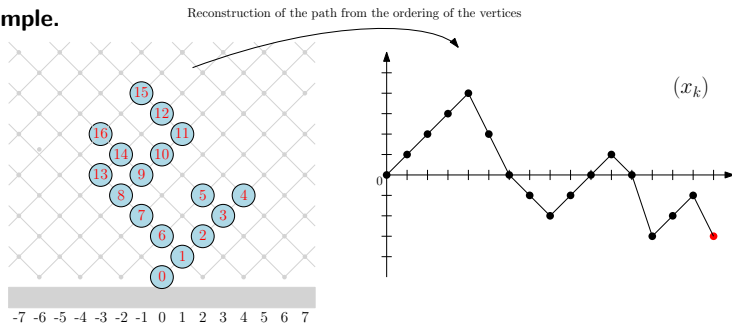
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Example.



$$u \text{ precedes } v \iff (u \text{ lifts } v) \text{ or } (\text{neither lifts the other and } x(u) > x(v)).$$

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*Pyramids are in **bijection** with paths (x_k) such that*

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Remarks

- $|A| = |x|$: this is a bijection preserving the size.
- By restriction : Bijection between half-pyramids and non-negative paths.

Now : how to induce the uniform measure on these constrained paths?

The animal walk

- Let (X_n) be i.i.d. random variables with law

$$\mathbb{P}(X_n = k) = \frac{2^k}{3} \mathbf{1}_{\{k \in \mathbb{Z}_-^* \cup \{1\}\}},$$

that is, $X_n = +1$ with prob. $2/3$, or $X_n = -\text{Geom}_{\mathbb{N}}(1/2)$ with prob. $1/3$.

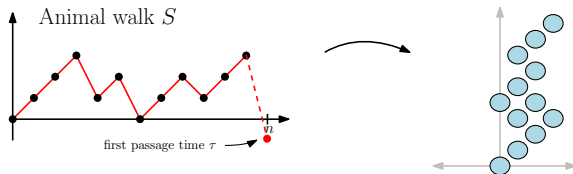
- The **animal random walk** is defined by $S_0 = 0$ and

$$S_n = X_1 + \cdots + X_n.$$

It is centered, with finite variance, and we set: $\tau = \inf\{n : S_n < 0\}$.

Proposition

A random uniform half-pyramid with n vertices can be sampled from an excursion of the animal random walk $(S_0, \dots, S_{\tau-1})$ conditioned on $\{\tau = n\}$.



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Proof.

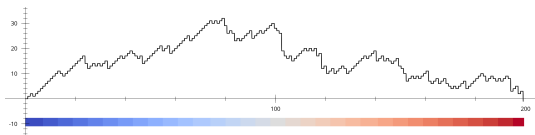
Given a path $0 = x_0, x_1, \dots, x_{n-1}$ with $x_i \geq 0$, $x_{i+1} - x_i \in \mathbb{Z}_-^* \cup \{1\}$,

$$\begin{aligned} \mathbb{P}(S_0 = x_0, \dots, S_{n-1} = x_{n-1}, S_n < 0) &= \sum_{x_n \leq -1} \prod_{i=1}^n \mathbb{P}(X_i = x_i - x_{i-1}) \\ &= \sum_{x_n \leq -1} \prod_{i=1}^n \frac{2^{x_i - x_{i-1}}}{3} = \sum_{x_n \leq -1} \frac{2^{x_n - x_0}}{3^n} = \frac{1}{3^n}. \end{aligned}$$

The Boltzmann half-pyramid

Definition

The Boltzmann half-pyramid (BHP) is the random directed animal constructed from a free positive excursion $(S_0, S_1, \dots, S_{\tau-1})$ of the animal random walk.

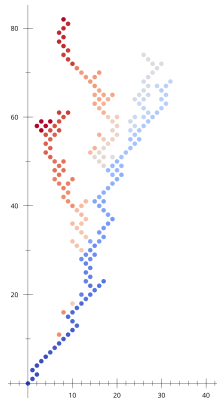


► BHP = critical half-pyramid: it is a.s. finite and gives probability $3^{-|A|}$ to any half-pyramid A . (g.f. at its critical point)

► The BHP admits the following recursive GW decomposition, where each possibility has probability $1/3$:

$$\text{BHP} = \left(\text{BHP}, \text{BHP}, \text{BHP} \right)$$

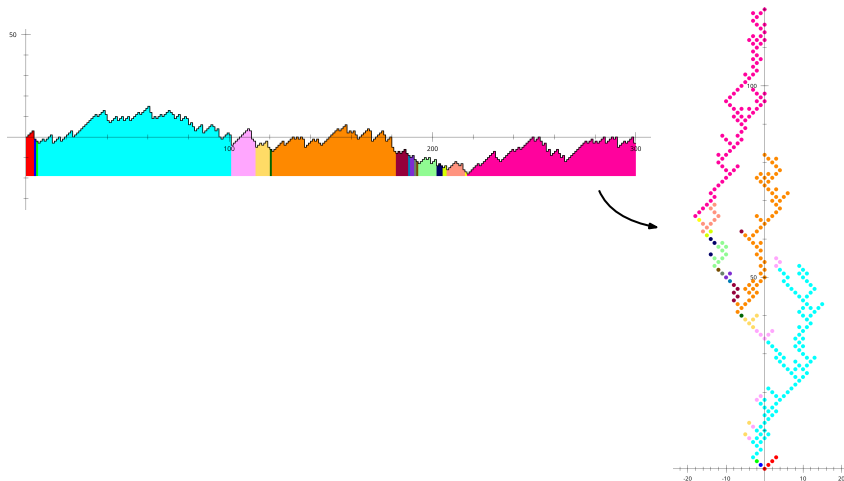
The diagram shows a green trapezoid (half-pyramid) on the left, followed by an equals sign, and then three green trapezoids in parentheses, each preceded by a small horizontal line segment. This represents the recursive decomposition of a half-pyramid into three smaller half-pyramids.



Local limit: the uniform infinite pyramid (UIP)

Theorem (Construction of the UIP)

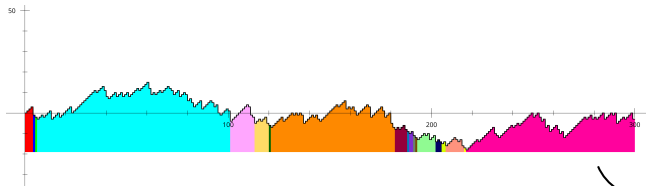
The local limit, rooted at 0, of a uniform pyramid with size n exists and is constructed by piling up i.i.d. BHPs rooted at $0, -1, -2, \dots$ in this order.



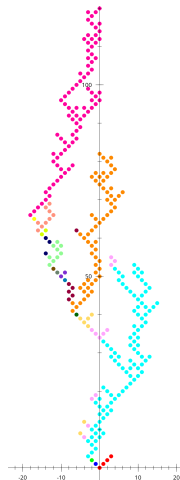
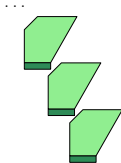
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► Schematically, in terms of the BHP, the UIP writes:



Local limit: the uniform infinite half-pyramid (UIP+)

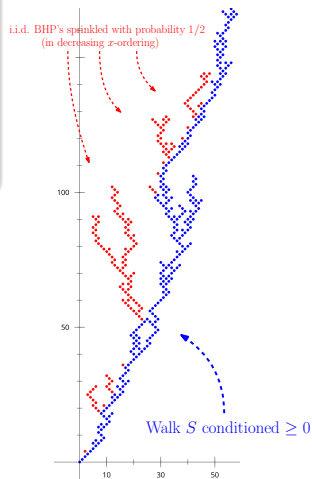
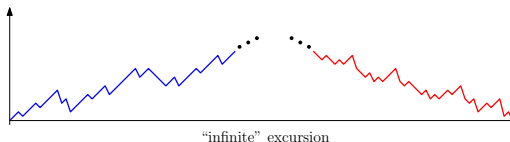
Theorem (Construction of the UIP+)

The local limit, rooted at 0, of a uniform half-pyramid with size n exists and is obtained:

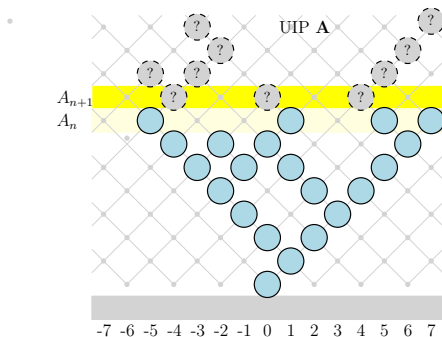
- by piling up the *animal walk conditioned to stay non-negative*.
- then by adding on top of it *i.i.d. BHPs* rooted at each integer $\in \mathbb{Z}^+$ with probability $1/2$, in decreasing order of x -coordinate.

► It is also the limit of BHPs when their size tends to ∞ .

- **Blue part:** beginning of the excursion.
- **Red part:** end of the excursion.



- Heaps of pieces constructions are conceptually nice, but the piling operation is very irregular.
- Now we ask about the evolution of the layers, and more precisely about the marginal law of the first n layers of the UIP $\mathbf{A}$, the BHP $\mathbf{A}^\circ$, and the UIP+ $\mathbf{A}^+$.



Marginals of the UIP

Let $L = \{x_1 < \dots < x_k\}$ and define

$$\eta(L) = \begin{cases} \prod_{i=1}^{k-1} (x_{i+1} - x_i - 1), & k \geq 2, \\ 1, & k = 1. \end{cases}$$

A sequence of layers $(L_0, \dots, L_n)$ is **admissible** if

$$L_{i+1} \subset (L_i - 1) \cup (L_i + 1), \quad 0 \leq i \leq n-1.$$

Theorem (UIP marginal)

Fix n . For every admissible sequence $L_0 = \{0\}, L_1, \dots, L_n$, the restriction $[\mathbf{A}]_{\leq n}$ of the UIP $\mathbf{A}$ to the first $n+1$ layers satisfies:

$$\mathbb{P}([\mathbf{A}]_{\leq n} = (L_0, \dots, L_n)) = \eta(L_n) 3^{-\sum_{i=0}^{n-1} |L_i|}.$$

Hence, for the UIP, the layers form a Markov chain with transition kernel

$$Q(L, L') = \frac{\eta(L')}{3^{|L|} \eta(L)} \mathbf{1}_{\{L' \subset (L-1) \cup (L+1)\}}.$$

Marginals of the BHP and UIP+

Theorem (UIP+ and BHP marginals)

Fix n . For every admissible sequence $L_0 = \{0\}, L_1, \dots, L_n$, the UIP+ $\mathbf{A}^+$ satisfies

$$\mathbb{P}([\mathbf{A}^+]_{\leq n} = (L_0, \dots, L_n)) = \frac{1}{2} (\min L_n + 1)(\max L_n + 2) \eta(L_n) 3^{-\sum_{i=0}^{n-1} |L_i|} \mathbf{1}_{\{L_0, \dots, L_n \subset \mathbb{Z}_+\}},$$

and the BHP $\mathbf{A}^\circ$ satisfies

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Remarks

- UIP+ and (on the non extinction event) BHP are positive-layer biases of the UIP marginal law.
- They provide us with martingales for the UIP.

UIP martingales

Let $\mathbf{A} = (A_n)_{n \geq 0}$ be the UIP sliced by layers. Set

$$m_n = \min A_n, \quad M_n = \max A_n, \quad \Delta_n = M_n - m_n, \quad \mathcal{F}_n = \sigma(A_0, \dots, A_n).$$

for the min, max and width of the n -th layer (beware, not *autonomous Markov*)

Corollary: UIP martingales

$$\mathbb{E}[M_{n+1} \mid \mathcal{F}_n] = M_n + \frac{1}{3^{|A_n|} \eta(A_n)}, \quad M_n + m_n \text{ is a martingale,}$$

and also,

$$\mathbb{E}[\Delta_{n+1} \mid \mathcal{F}_n] = \Delta_n + \frac{2}{3^{|A_n|} \eta(A_n)}.$$

Besides,

$$(m_n + 1)(M_n + 2) \text{ is a martingale.}$$

Proof idea.

Compare the expressions of the UIP, BHP and UIP+ kernels.

Sausaging of the UIP

Define the pinching time:

$$T = \inf\{n > 0 : \Delta_n = 0\} = \inf\{n > 0 : |A_n| = 1\}.$$

Theorem (Sausaging property)

For the UIP,

$$\mathbb{P}(T < \infty) = 1.$$

Moreover,

$$\mathbb{E}[T] = \infty, \quad \mathbb{E}\left[\max_{n < T} \Delta_n\right] = \infty.$$

Proof idea (first statement)

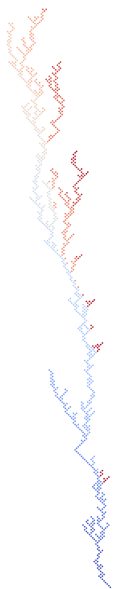
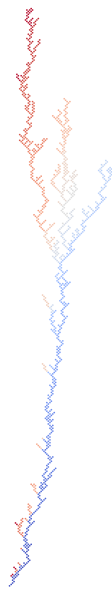
The width process satisfies

$$\mathbb{E}[\Delta_{n+1} - \Delta_n \mid \mathcal{F}_n] = \frac{2}{3^{|A_n|}\eta(A_n)} \leq \frac{2}{9(\Delta_n - 1)} \quad (\Delta_n > 1),$$

Together with the variance lower bound, $\mathbb{E}[(\Delta_{n+1} - \Delta_n)^2 \mid \mathcal{F}_n] \geq \frac{16}{9}$, this implies

$$Y_n = \log(1 + \Delta_n)$$

is a supermartingale outside a finite set. Hence Δ_n returns to small widths a.s., and irreducibility then gives an a.s. finite pinching time.

- 
- 
- **Uniformly** rooted local limit.
 - Does a large directed animal typically contain **macroscopic** loops?
 - **Height** of a directed animal with size n :

$$H_n \stackrel{?}{\asymp} n^\alpha.$$

(Open for a long time, $\alpha \approx 0.817$, numerics only.)

- **Scaling limit** for large directed animals, more generally.
- **Universality.** Tackle other solvable 2D directed lattices.

The slide features two decorative, fractal-like lines on the left and right sides. These lines are composed of many small, connected segments in red, orange, and blue, creating a complex, branching pattern that resembles a stylized tree or a mathematical fractal. They are positioned vertically, framing the central text.

Thank you for your attention!

Branching-annihilating particle system

Consider a system with  and  particles such that:

- 1 Each particle (at i) reproduces independently, creating particles at $i-1, i, i+1$ s.t.

$$\mathbb{P}\left(\begin{array}{c} \text{light blue} \\ \diagup \quad \diagdown \\ \text{light blue} \end{array}\right) = \mathbb{P}\left(\begin{array}{c} \text{light blue} \\ \diagdown \quad \diagup \\ \text{light blue} \end{array}\right) = \mathbb{P}\left(\begin{array}{c} \text{light blue} \quad \text{red} \quad \text{light blue} \\ \diagup \quad \diagdown \quad \diagup \\ \text{light blue} \end{array}\right) = \frac{1}{3} \quad \text{and} \quad \mathbb{P}\left(\begin{array}{c} \text{red} \\ \diagup \quad \diagdown \\ \text{red} \end{array}\right) = 1$$

- 2 Particles of opposite colors annihilate when they collide.



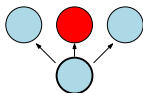
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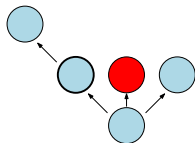
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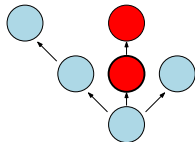
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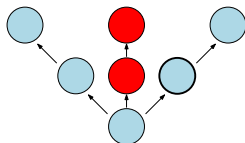
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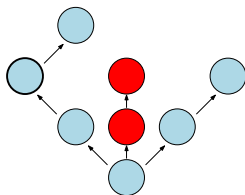
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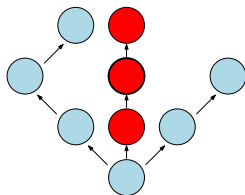
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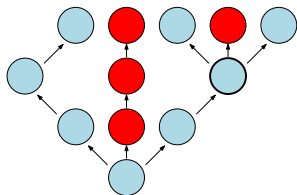
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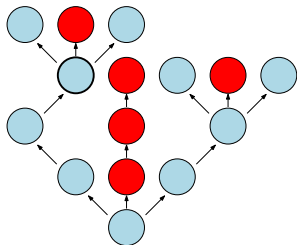
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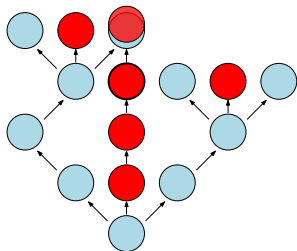
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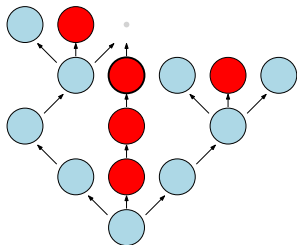
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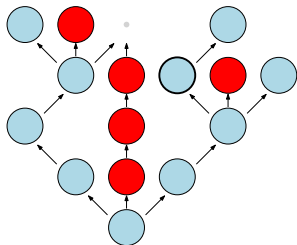
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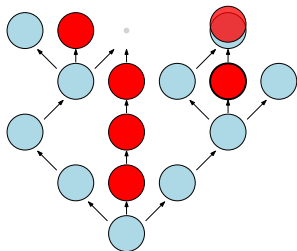
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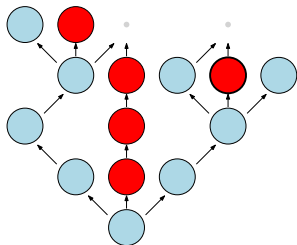
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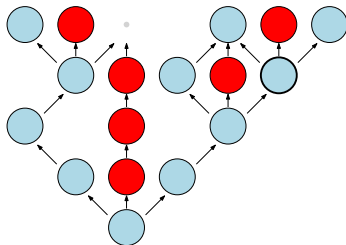
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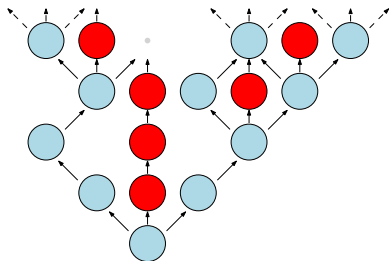
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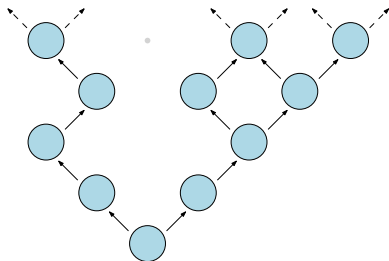
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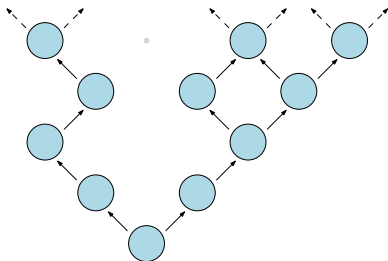
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- function of a Markov process is Markov
 $\implies$ Intertwining of kernels.
 $\implies$ Intertwining of red/blue particles.
- The red particle can also move!
- The long-range interaction between blue vertices is mediated by the “invisible red particles”.
- Reminiscent of
 - ▶ Dyson’s Brownian motion
 - ▶ Pitman’s theorem.

↖ This is the UIP !